

EXPERIMENT 15: DECISION TREE

Aim

To write a Python program to implement a **Decision Tree classification algorithm**.

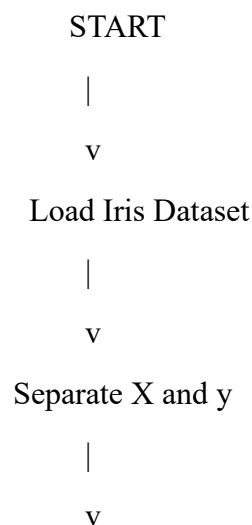
Objective

- To understand supervised machine learning.
- To build a classification model using a Decision Tree.
- To train the model using training data.
- To predict the class of test data.
- To calculate the classification accuracy.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Separate features X and target y.
4. Divide the dataset into training and testing sets.
5. Create a Decision Tree classifier.
6. Train the classifier using training data.
7. Predict the classes of test data.
8. Compare predicted and actual values.
9. Calculate accuracy.
10. Display the prediction and accuracy.

Flowchart



Split Train/Test

|

v

Create Decision Tree

|

v

Train the Model

|

v

Predict Test Data

|

v

Calculate Accuracy

|

v

Display Result

|

v

END

Python Program

Writing

Decision Tree Classification

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Load Iris dataset
iris = load_iris()
```

```
X = iris.data
y = iris.target

# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

# Create Decision Tree model
model = DecisionTreeClassifier(random_state=42)

# Train the model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Actual values:")
print(y_test)

print("Predicted values:")
print(y_pred)

print("Accuracy:", accuracy)
```

```
# Sample prediction
sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(sample)

print("Sample Prediction:",
      iris.target_names[prediction][0])
```

Sample Output

Actual values:

```
[1 0 2 1 1 0 1 2 1 1 ...]
```

Predicted values:

```
[1 0 2 1 1 0 1 2 1 1 ...]
```

Accuracy: 1.0

Sample Prediction: setosa

Note: The exact accuracy can vary depending on the train/test split and library version.

Result

Thus, the **Decision Tree classification algorithm** was successfully implemented using the **Iris dataset** and its classification accuracy was calculated.